Week 2 Update

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Weekly progress meeting with Professor Torquato.
Main topics:

- Recap of last week
- New patterns analyzed (Classes II, III)
- The Bombieri-Taylor result
- Updated ranking

Quick Recap: Last Week

Last week:

- Variance code validated (Poisson benchmark, 1.1% error)
- Three metallic-mean chains: Fibonacci, Silver, Bronze
- All gave $\alpha = 3$ as expected

Pattern	α (measured)	α (expected)	$\bar{\Lambda}$
Fibonacci	3.05	3	0.200
Silver	2.99	3	0.250
Bronze	2.99	3	0.282

This week: Cover all three hyperuniformity classes and find something in the $1 < \alpha < 2$ gap.

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This week: Cover all three hyperuniformity classes and find something in the $1 < \alpha < 2$ gap.

Quick reminder of where we left off.
The metallic means all give $\alpha = 3$ because of the 2x2 matrix constraint: $|\lambda_2| = 1/|\lambda_1|$.
Goal this week was to expand beyond just $\alpha = 3$.

Class II: Period-Doubling

The Period-Doubling chain:

$$S \to L, \quad L \to LSS \qquad \Rightarrow \qquad M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix}$$

Matrix gives $\lambda_1 = 2$, $|\lambda_2| = 1$

So: $\alpha = 1 - \dfrac{2\ln(1)}{\ln(2)} = 1$

Results:

- Variance grows like $\ln R$ (not bounded, not power-law)
- $\bar{\Lambda}$ diverges — can't use it to rank this pattern
- This is the boundary case: $\alpha = 1$ exactly
- Class II confirmed

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Period-Doubling is the boundary case: $\alpha = 1$ exactly.
$|\lambda_2| = 1$ means $\ln(1) = 0$ in the formula.
First Class II example — variance grows logarithmically.

Class III: 0222 Chain

The 0222 chain:

$$S \to LL, \quad L \to SSLL$$

Here $|\lambda_2| = \sqrt{5} - 1 > 1$
So: $\alpha \approx 0.64$

Results:

- Variance grows like $R^{0.36}$
- $\bar{\Lambda}$ diverges
- Class III confirmed

Now have examples of all three classes.

The three classes:

Class	α	$\sigma^2(R)$
I	> 1	bounded
II	$= 1$	$\sim \ln R$
III	< 1	$\sim R^{1-\alpha}$

The exponent $1 - 0.64 = 0.36$ matches the measured value.

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0222 has $|\lambda_2| > 1$, which gives $\alpha < 1$.
The power-law growth $R^{0.36}$ matches the prediction $R^{1-\alpha}$.
All three classes now covered.

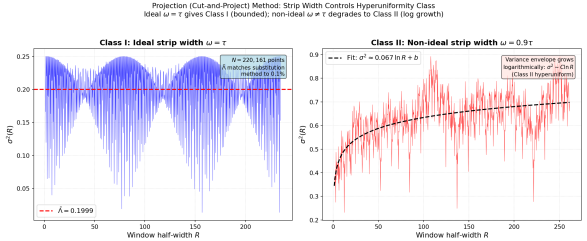
URL Verification

Tested the perturbed lattice formula:

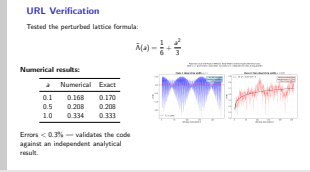
$$\bar{\Lambda}(a) = \frac{1}{6} + \frac{a^2}{3}$$

Numerical results:

a	Numerical	Exact
0.1	0.168	0.170
0.5	0.208	0.208
1.0	0.334	0.333



Errors $< 0.3\%$ — validates the code against an independent analytical result.



URL = uncorrelated random lattice (independent displacements).
Good validation because we have the exact formula.
Always $\alpha = 2$ regardless of displacement amplitude a.

Stealthy Patterns

Analyzed stealthy configurations provided by Sam.

Idea: Force $S(k) = 0$ for all $|k| < K^*$, not just at $k = 0$.

Results ($N = 2{,}000$, $\sim$4000 configs each):

χ	$\bar{\Lambda}$ (measured)	$\bar{\Lambda}$ (predicted)	α
0.1	1.02	1.01	2.0
0.2	0.53	0.51	2.0
0.3	0.36	0.34	2.0

Key observation: $\bar{\Lambda} \approx 1/(\pi^2\chi)$ — good agreement with theory.

Higher χ (larger exclusion zone) $\Rightarrow$ better hyperuniformity, approaching lattice as $\chi \to \infty$.

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Data from another grad student (CCO format, 4000 configs per χ).
χ controls the size of the exclusion region: $K = 2\pi\chi$.
Theory predicts $\bar{\Lambda} \approx 1/(\pi^2\chi)$.
Larger exclusion zone = better suppression of fluctuations.
$\alpha = 2$ is theoretical (can't extract via spreadability due to exponential decay).

Filling the $1 < \alpha < 2$ Gap

There was a gap in the α coverage:

α	Class	Examples
3	I	Fibonacci, Silver, Bronze
2	I	URL, Stealthy
$1 < \alpha < 2$	I	No quasicrystal
1	II	Period-Doubling
< 1	III	0222

The issue: all 2×2 metallic-mean matrices give $\alpha = 3$ because $|\lambda_2| = 1/|\lambda_1|$.

Solution: the Bombieri-Taylor cubic substitution (1986 paper, p. C3-21).

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Solution: the Bombieri-Taylor cubic substitution (1986 paper, p. C3-21).

The 2x2 constraint: $\det(M) = \pm1$ forces $|\lambda_2| = 1/|\lambda_1|$.
This always gives $\alpha = 1 + 2 = 3$.
Need a 3x3 matrix to break this constraint.

Bombieri-Taylor Cubic Substitution

From the 1986 paper (p. C3-21):

Three-letter alphabet:

$$a \to aac,\ b \to ac,\ c \to b$$

Matrix:

$$M = \begin{pmatrix} 2 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Eigenvalues: $\theta_1 = 2.247$, $|\lambda_2| = 0.802$

Predicted:

$$\alpha = 1 - \frac{2\ln(0.802)}{\ln(2.247)} = \mathbf{1.545}$$

Implementation:

- Extended code for 3-letter alphabets
- Generated 9.5 million points
- Computed variance and $\bar{\Lambda}$

Result: $\bar{\Lambda} = 0.377$, Class I confirmed.

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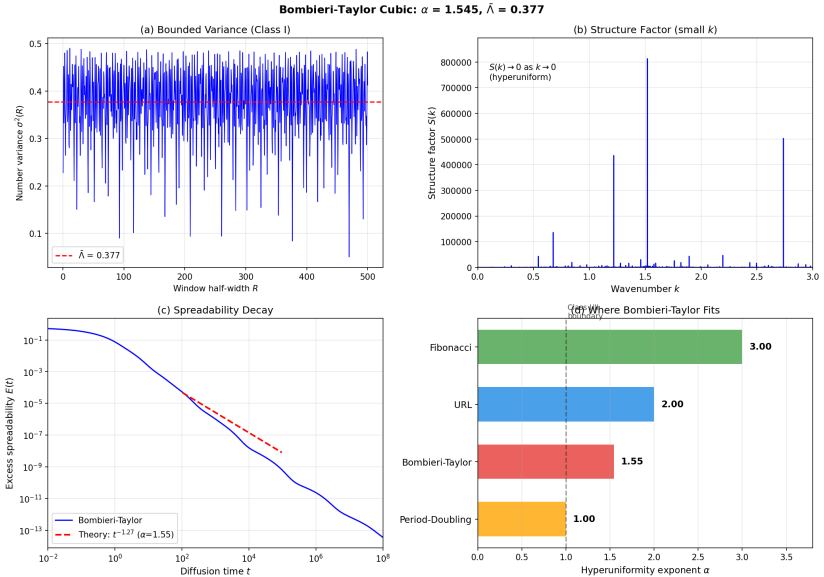
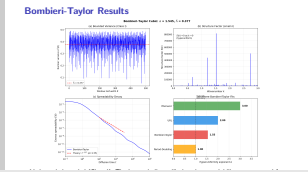
Result: $\bar{\Lambda} = 0.377$, Class I confirmed.

Key: $|\lambda_2| = 0.802 \neq 1/\theta_1 = 0.445$.
The 3x3 matrix breaks the constraint.
θ_1 is a Pisot number (conjugates all < 1).

Bombieri-Taylor Results



Variance is bounded (Class I). The log-periodic oscillations in spreadability make power-law fitting unreliable for quasicrystals. The eigenvalue formula is exact; spreadability fit is not.



Panel (a): Bounded variance confirms Class I. Oscillates between 0.05–0.49, mean = 0.377.
Panel (b): Structure factor at small k. Shows $S(k) \to 0$ as $k \to 0$ (hyperuniform). Bragg peaks are characteristic of quasicrystals.
Panel (c): Spreadability decay. The log-periodic oscillations make power-law fitting unreliable for quasicrystals. The eigenvalue formula is exact; spreadability fit is not.
Panel (d): Shows Bombieri-Taylor fills the gap between Period-Doubling ($\alpha = 1$) and URL ($\alpha = 2$).
Confidence: $\alpha = 1.545$ is exact (analytical from eigenvalues). $\bar{\Lambda} = 0.377$ is numerical average of bounded variance.

Updated Ranking

Here's where everything stands now:

Pattern	Class	α	$\bar{\Lambda}$
Lattice	I	∞	0.167
URL ($a = 0.1$)	I	2.0	0.168
Stealthy ($\chi = 0.3$)	I	2.0	0.36
Fibonacci	I	3.0	0.200
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URL ($a = 1.0$)	I	2.0	0.333
Bombieri-Taylor	**I**	**1.545**	**0.377**
Period-Doubling	II	1.0	diverges
0222	III	0.64	diverges

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Lattice | I | ∞ | 0.167
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Bombieri-Taylor | I | 1.545 | 0.377
Period-Doubling | II | 1.0 | diverges
0222 | III | 0.64 | diverges

26 patterns total now (counting different χ and a values).
Bombieri-Taylor fills the gap between $\alpha = 1$ and $\alpha = 2$.
First quasicrystal with $1 < \alpha < 2$.

Summary

This week:

- Added Class II (Period-Doubling, $\alpha = 1$)
- Added Class III (0222, $\alpha = 0.64$)
- Verified URL formula to $< 0.3\%$
- Analyzed stealthy data ($\bar{\Lambda} \approx 1/(\pi^2\chi)$)
- Implemented Bombieri-Taylor cubic ($\alpha = 1.545$)

Main result: Filled the $1 < \alpha < 2$ gap with a quasicrystal.

Questions:

- Are there other cubic/quartic substitutions to try?
- Should 2D generalizations be next?

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- Should 2D generalizations be next?

Total patterns analyzed: 26
Key accomplishment: Bombieri-Taylor fills the gap.
Possible next steps: other cubic substitutions, 2D/3D.

References

- Torquato & Stillinger (2003), Phys. Rev. E 68, 041113
- Bombieri & Taylor (1986), J. Physique Colloques 47, C3-19
- Oğuz et al. (2017), Phys. Rev. B 95, 054119
- Torquato (2021), Phys. Rev. E 104, 054102

References

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Main references used this week.